

Tunnel Field-Effect Transistors

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Abstract – Tunnel Field-Effect Transistors (TFETs) are emerging semiconductor devices that use quantum tunneling for carrier injection. They show promising results in overcoming the steep subthreshold swings below the 60 mV/decade limit of conventional MOSFETs. By leveraging band-to-band tunneling (BTBT) instead of thermionic emission, TFETs can operate at much lower supply voltages while maintaining low leakage currents, making them desirable for ultra-low-power electronics [1]. This report provides an overview of TFETs, including their historical development, operating principles, device applications, and the current state of industrial use and market prospects. It will also explain the steep switching limit and how this phenomenon limits circuit potentials and in turn it will also cover how TFETs are able to overcome this steep switching threshold for traditional MOSFETs via BTBT. While TFETs offer significant power advantages, they face challenges such as lower on-currents, thus far limiting their commercial adoption. Nonetheless, ongoing research and recent breakthroughs indicate a strong potential for TFETs in future low-power and energy-efficient electronics [2].

INTRODUCTION

Power dissipation and the inability to further scale supply voltages have become critical challenges in modern complementary Metal-Oxide-Semiconductor (CMOS) technology, a technology that is used extensively in ICs, largely due to the fundamental subthreshold swing (SS) limit of ~ 60 mV/decade at room temperature in MOSFETs. This limit, which is the result of the thermionic emission process, means that lowering the threshold voltage to reduce operating voltage inevitably leads to exponentially rising leakage currents [1].

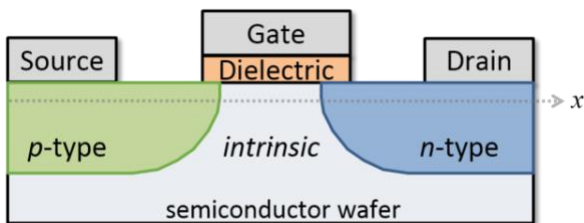


Figure 1: p-i-n structure of a n-type TFET. The structure of a p-type TFET is similar with the n and p-type semiconductors swapped between the source and the drain [4].

As transistor scaling reached nanoscale dimensions in the late 1990s, the need for new transistor mechanisms with steeper switching characteristics became apparent. TFETs emerged as a possible solution. A TFET operates more like a gated p-i-n diode, where i represents an intrinsic region which would typically be undoped or very lightly doped [3], as shown in figure 1.

Despite their promising results in being very power efficient, they are yet to be widely used in industrial applications as they are still being researched and developed to overcome limitations.

HISTORY

The concept of utilizing tunneling for transistor action dates back several decades. The first TFET-like device concept was introduced in 1965 by Hofstein and Warfield, who proposed an “insulated gate tunnel junction triode” [5]. This early idea essentially envisioned using gate control to modulate tunneling across a reverse-biased junction. Although its structure was very different, it laid the conceptual groundwork for TFETs.

For the next 40 years there were no significant breakthroughs with TFETs that made them more appealing than existing transistors during the same period. The most significant breakthrough came in the early 2000s. In 2004, Joerg Appenzeller and colleagues at IBM experimentally showed a gated carbon nanotube transistor exhibiting swing below 60 mV/decade, confirming that TFETs could indeed beat the MOSFET thermal limit [6].

This result was pivotal as it prompted a wave of research into various TFET designs and materials. By the late 2000s, multiple groups had demonstrated TFET operation in different material systems, including silicon, germanium, and III-V compound semiconductors. Researchers even began introducing techniques like heterojunction source-channel interfaces to enhance tunneling probability [7] to improve drive current.

However, there have also been challenges. TFET on-currents have often been relatively low, especially in silicon-based TFETs, and material integration is complex. In the last few years, demonstrations have begun including hybrid TFET-MOSFET circuits by IBM, indicating industry interest in implementing TFETs for low-power functionalities [8].

DEVICE OPERATION

At its core, a TFET is a gated p-i-n diode operated in reverse-bias. Unlike traditional MOSFETs, where the carriers need the minimum energy to overcome the band “wall”, in TFETs the carriers essentially “phase” across the band “wall”.

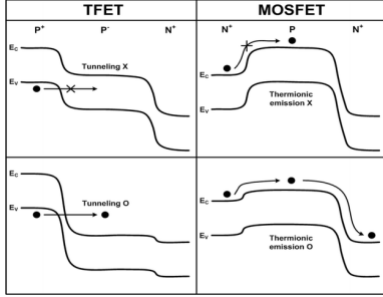


Figure 2: Energy band diagram of TFET vs MOSFET in operation [9].

Figure 2 illustrates this idea really clearly, the top 2 diagrams for TFET and MOSFET show the electrons unable to cross the energy band barrier, which can be visualized as an energy wall. Typically, in a MOSFET for the electrons to cross from one end of the diode to the other end, there is a gate voltage applied to push the conduction band down to lower the energy barrier between the source and channel. This enables thermionic emission as the electrons can hop over the barrier with a small amount of energy. Since electrons like to “sink”, they continue to move to the drain, resulting in the flow of current. TFETs on the other hand need a strong gate voltage to push the conduction band of the channel below the valence band of the source, which is why it is essentially operated in “reverse-bias”. When a strong gate voltage is applied, at the interface between the p-type and intrinsic semiconductors, the band structure is bent so much such that across the p-i junction, the conduction and valence band are right next to each other. At these small spatial widths (~1-3 nm) quantum effects begin to dominate and through a process known as quantum tunnelling, the electrons are then able to tunnel through the energy barrier. So essentially, the TFET turns “on” by enabling BTBT at the source channel interface rather than lowering the thermal barrier as in a MOSFET.

Unlike MOSFETs, TEFTs have 2 main modes of operation: Cutoff region and Tunneling region. This is because the current flowing through a TFET is dependent on the tunneling probability and not the carrier drift velocity [9].

Cutoff Region ($V_{GS} = 0$):

When there is no gate source voltage, the energy band diagram remains unchanged and resembles that of the top left graph in figure 2. Therefore, the carriers are unable to move through the transistor since the spatial width between the conduction and valence band is too big. Since quantum tunneling only occurs at very small lengths, the BTBT becomes almost negligible since the width between the conduction and valence bands are relatively large in this

operating region. As a result, the only leakage that occurs is mostly from thermally generated stimuli.

Tunneling Region ($V_{GS} > V_{TH}$):

As the gate-source voltage increases from 0V, the energy band begins to bend at the source-channel interface. This bending reduces the width of the tunneling barrier between the valence band and conduction band at the interface. As a result, the BTBT rate slowly increases as the gate-source voltage increases. Eventually there will be a high enough BTBT rate, allowing a significant number of carriers to pass, leading to significant current flow through the transistor. This is the threshold voltage, and as the gate-source voltage keeps increasing, the band structure continues to bend, further increasing the tunneling window region. As a result, the drain current also increases. Unlike MOSFETs where saturation occurs due to channel pinch-off, TFETs exhibit a different behavior. As the drain-source voltage gets very large, at some point the increase in drain current will diminish due to the limits of how much the band structure can bend. At some point the electric field required to bend the band structure would be too large. This limit eventually means the tunneling window does not get larger, which results in a quasi-saturation phenomenon [10].

It is due to how the TFET operates that it is able to overcome the ~60mV/dec steep switching limit, which describes how fast the transistor can transition from the off state. With the increased V_{GS} , and as the energy barrier width decreases, the tunneling probability exponentially increases. This proportional relationship is why its steep switching limit is <60mV/dec.

To better visualize the difference the drain current for TFET will be computed using an approximate formula [11]. The total drain current can be obtained by integrating the charge generated throughout the whole device. Below the generation rate will be represented with $G(E)$. For simplification, it will be assumed that the variation along the x and y direction can be neglected.

$$I_D = q \int G(E) dV = qWL \int G(E) dz$$

As for the generation rate, it will be assumed that the semiconductor is in a uniform electric field. With this assumption the Kane’s Model to calculate generation will be used since it is currently the most popular model in calculating the generation rate for the BTBT process [12]. The Kane’s Model is given by:

$$G(E) = A \frac{E^D}{\sqrt{E_g}} e^{\left(\frac{-BE_g \left(\frac{3}{2} \right)}{E} \right)}$$

Where E is the electric field, A and B are material dependent constants, D is a parameter that depends on the

tunneling mechanism. Since evaluating this integral would not be possible, an approximated equation will be used [11]. The approximate drain current can be given by:

$$I \approx \frac{WLA\sqrt{q}}{BE_g^{\frac{3}{2}}} \left(\frac{E_g N_a}{2\epsilon_s} \right)^{D/2} e^{Bq\sqrt{2\epsilon_s E_g}/\sqrt{qN_a} \left(\sqrt{\delta\psi} - \sqrt{\frac{E_g}{q}} \right)} \sqrt{\delta\psi}$$

Where $\delta\psi$ is a function of the gate voltage. This approximated was selected as it yields results similar to numerically evaluating the integral [11]. Using this equation and the known current equations of a MOSFET:

1. Linear region:

$$I_D = \mu_n C_i \frac{Z}{L} \left[(V_G - V_T)V_D - \frac{V_D^2}{2} \right]$$

2. Saturation region:

$$I_D = \mu_n C_i \frac{Z}{2L} (V_G - V_T)^2$$

The graphs for the current of the MOSFET and TFET can be computed using MATLAB, while using a similar footprint for both transistors.

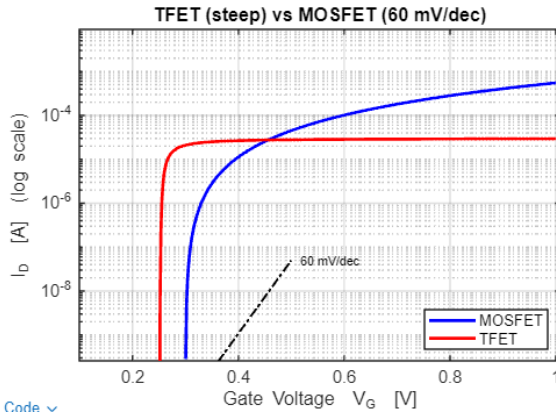


Figure 3: TFET vs MOSFET drain current for intrinsic Si properties, $V_T=26$ mV, $W = 1e-6$ m, $L = 40e-9$ m, $E_g = 0.66$ eV, $N_a = 5e25$ m⁻³, $A = 4e19$ (m⁻²s⁻¹), $B = 23e9$ (V⁻¹m⁻¹), $V_{on} = 0.2$ and $V_{th} = 0.35$.

From figure 3, it becomes evident how much faster TFETs are capable of switching compared to MOSFETs at $T = 300$ K. This would hold true for a large range of temperatures since TFETs work on the principle of quantum tunneling and not thermionic process. This is also evident through the equations, since V_T and μ_n are both temperature dependent, while in a TFET, the exponent is the dominant term, since the dimensions will be in the magnitude of nano, and the square root of the parameters will also be very tiny, so its dependance on temperature is incredibly small.

These properties of small steep switching limit, small temperature dependance and smaller threshold voltage make TFETs incredibly appealing for their performance efficiency. However, figure 3 also identifies a major limitation to TFETs

compared to MOSFETs. The I_D curve for the MOSFET continues to increase and has still not reached the saturation point, whereas the TFET has already reached its quasi-saturation point around the 0.4V gate voltage. Thus, for amplification purposes, TFETs are not ideal. This can be verified by checking the gain formula. Since the gain A_v is given by the equation:

$$A_v = g_m \cdot r_o$$

Where g_m is given by:

$$g_m = \frac{dI_D}{dV_{GS}}$$

Even though the g_m of TFETs are high at initial voltages, at higher voltages the g_m diminishes, meaning the transconductance of the transistor is only high for the small range of voltages required to turn on the transistor. This does not make it very practical. This analysis is further supported by a journal written in 2011 comparing the power and performance for the TFET and MOSFET [13].

From the analysis on the operation of TFETs it is clear that it is more suited for low power circuits that want to prioritize power efficiency. Examples could include chips that would be used in large servers or for training AI models. In addition CPUs and GPUs would also benefit from faster switching times.

However, despite these advantages practical TFETs exhibit subthreshold swings worse than theoretical predictions. This is due to the scattering and trap assisted tunneling effects (TAT). This is a process where a localized energy level, known as a trap, that lies in the energy band gap can capture a carrier and release it. This two-step process allows TAT to occur at lower electric fields which in turn leads to a leakage current. This effect has both an electric field and temperature dependance, magnifying this effect [7].

As a result, the presence of TAT will degrade the subthreshold swing, since even when the TFET is nominally off, traps can facilitate some current. This would result in the steep I-V curve from figure 3 to look similar except with a tail at the subthreshold voltage values.

If the density of trap states is high, TAT can dominate the OFF-state and subthreshold behavior. This is believed to be one reason many experimental TFETs show measured SS in the 100–200 mV/dec range instead of the expected <60 mV/dec [14].

DEVICE APPLICATIONS

One key application area where TFETs are sought after is digital logic circuits for power-critical systems. By using TFETs in place of MOSFETs, logic gates could, in theory, operate at significantly lower voltages (around 0.3–0.5 V instead of 1 V) and still achieve acceptable on-currents,

thereby reducing dynamic power consumption by large factors.

TFET-based logic is also appealing for battery-powered and wearable devices, “always-on” sensors, and IoT nodes that spend much of their time in standby but require occasional active operation with very low energy budgets. This is because having a TFET constantly on would be incredibly energy efficient compared to MOSFETs, since they require less voltage to drive current [15].

TFETs are more likely to find use in moderate-speed analog or sensor interface circuits that prioritize power savings over bandwidth. For instance, a TFET-based source-follower or instrumentation amplifier in an IoT sensor node could significantly cut bias power. There has even been research into using TFETs for biosensors and sensing applications – the idea is that the TFET’s gated diode structure is highly sensitive to surface potential changes, which can be utilized in label-free biochemical sensing [16].

Overall, TFETs hold promise to significantly extend battery life and reduce heat in future electronics, assuming their integration challenges can be overcome. As more research still needs to be conducted to improve TFETs, they are yet to be implemented in large scale industrial purposes.

MARKET ANALYSIS

Despite TFETs not being implemented for large scale industrial purposes, according to market analysis reports, the TFET market, which includes research activities, was valued at around USD 984 million in 2023, and is projected to grow to about USD 2.07 billion by 2030 (an 11.2% CAGR) [17]. This suggests that TFETs are still emerging technology, that requires major breakthroughs as its growth for the next decade is relatively small compared to transistors like FinFETs.

Nonetheless, TFETs may experience a major “market boom” due to the advancements in AI and the ever growing concerns on the environmental impact data centers are beginning to pose.

CONCLUSION

Tunnel field-effect transistors represent a promising path toward ultra-low-power electronics by circumventing the fundamental thermal limit of MOSFETs. This report reviewed the TFET’s operating principle – how a gated p–n junction can exploit band-to-band tunneling to achieve steep subthreshold slopes – and examined in depth the device physics that govern its behavior.

Comparing TFETs to MOSFETs, it’s clear that TFETs have an edge in potential energy efficiency, with a much lower leakage and threshold voltage, but lag in current drive and maturity. Nevertheless, a desire for more power efficient and faster transistors keeps the interest in TFETs alive.

Applications in low-power logic, memory, and sensing could greatly benefit from TFET integration.

Industry has recognized this potential, with major semiconductor companies such as Texas Instruments and TSMC researching TFETs as a candidate for beyond-CMOS technology [18]. Market analysis predict a growing role for TFETs, particularly as demand for energy-efficient electronics shows no signs of stopping.

In conclusion, TFETs stand as potential successors to current transistors as they operate through a different process compared to conventional MOSFETs.

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